



Introduction

Modern software project management has become increasingly challenging due to complex task dependencies, resource constraints, changing requirements, and the need for efficient collaboration. This project presents an AI-powered Agile Project Management System that integrates Artificial Intelligence with Design and Analysis of Algorithms (DAA) to automate task prioritization, sprint planning, dependency analysis, resource allocation, and team assignment. By utilizing algorithms such as Greedy, Topological Sort, 0/1 Knapsack, Branch and Bound, Boyer-Moore-Horspool, Merge Sort, and Breadth-First Search (BFS), the system generates optimized execution plans, improves resource utilization, enhances decision-making, and enables faster, smarter, and more reliable project management for Agile software development.

Problem Definition

- Manual Project Planning:** Traditional project management relies heavily on manual task scheduling, prioritization, and dependency management, making the process time-consuming and error-prone.
- Inefficient Resource Allocation:** Assigning tasks without considering member skills, workload, and task dependencies often leads to poor resource utilization and reduced team productivity.
- Need for an Intelligent Solution:** An AI-powered project management system integrating DAA algorithms is required to automate task prioritization, sprint planning, dependency analysis, and optimal team assignment.

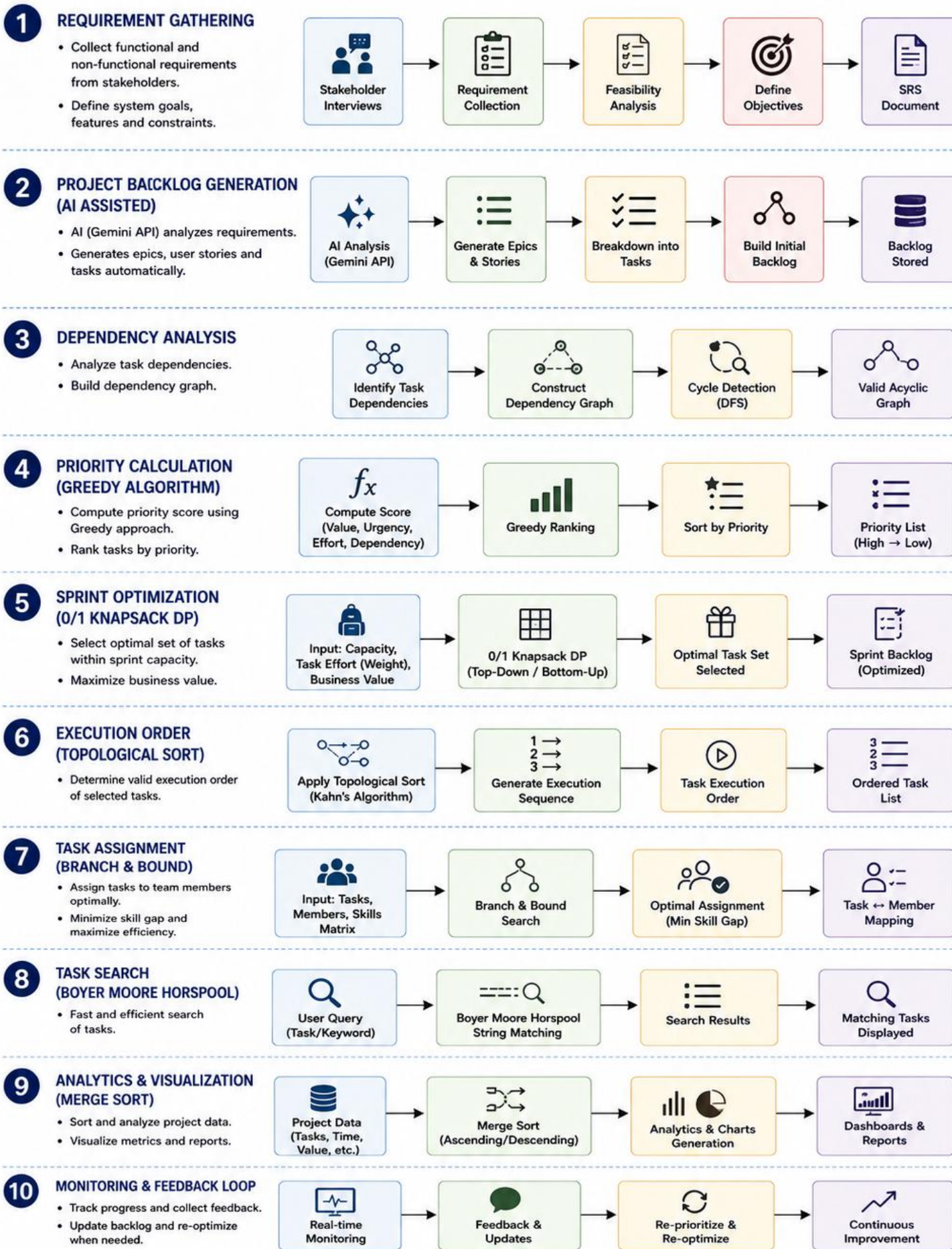
Objectives

- Develop an **AI-powered Agile Project Management System** using Design and Analysis of Algorithms (DAA).
- Automate **task generation, prioritization, and dependency management** using Artificial Intelligence and optimization algorithms.
- Optimize **sprint planning** by selecting the most valuable tasks within the available sprint capacity using the **0/1 Knapsack Algorithm**.
- Generate a **dependency-aware execution roadmap** using **Topological Sort** to ensure tasks are completed in the correct order.

Methodology

NEXUSFLOW – AI POWERED AGILE PROJECT MANAGEMENT SYSTEM

Methodology (Design and Analysis of Algorithms)



This methodology combines multiple Design and Analysis of Algorithms techniques to automate planning, optimization, assignment and tracking in an Agile project management environment.

Tools used

Frontend Technologies: React.js, TypeScript, Tailwind CSS, HTML5, CSS3.
Backend & Database: Node.js, Express.js, MongoDB, Mongoose, REST APIs, Socket.IO.
Development & Configuration: Visual Studio Code, Git & GitHub, Postman, npm, MongoDB Compass.
Artificial Intelligence & Project Planning: Google Gemini API, AI-based Project Decomposition, Task Generation, Sprint Recommendation, Dependency Analysis.
Design & Analysis of Algorithms: Greedy Algorithm, Topological Sort, 0/1 Knapsack, Branch and Bound, Merge Sort, Boyer-Moore-Horspool, Breadth-First Search (BFS), Depth-First Search (DFS) for intelligent task prioritization, execution planning, sprint optimization, resource allocation, dependency management, and efficient project workflow optimization.

Results and Discussions

- Successfully automated project planning, task prioritization, dependency management, sprint optimization, and team assignment using Artificial Intelligence integrated with Design and Analysis of Algorithms (DAA).
- Achieved efficient, explainable, and optimized Agile project management through intelligent task scheduling, resource allocation, dependency analysis, and real-time collaboration, making the system suitable for modern software development teams.

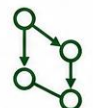


RESULTS



Optimized Sprint Planning

0/1 Knapsack algorithm selects the optimal set of tasks maximizing business value within the sprint capacity.



Correct Execution Order

Topological Sort generates a valid and efficient execution sequence respecting all task dependencies.



Optimal Task Assignment

Branch & Bound algorithm assigns tasks to the most suitable team members while minimizing the overall skill gap.



Fast and Accurate Search

Boyer Moore Horspool algorithm enables fast and efficient searching of tasks and backlog items.



Efficient Data Handling

Merge Sort efficiently sorts and organizes large datasets for analytics, reporting and visualizations.



Reliable Dependency Analysis

BFS identifies dependency-free tasks and detects cycles to ensure project integrity and feasibility.



All algorithms performed effectively with optimal time complexity, ensuring scalability, accuracy and efficiency.



DISCUSSION



Integrated Algorithmic Approach

The combination of multiple DAA techniques enables intelligent automation of planning, scheduling, assignment and tracking in Agile project management.



Performance & Efficiency

Algorithms like Knapsack DP and Branch & Bound provide high-quality optimal results within acceptable execution time, suitable for real-world projects.



Scalability

The system efficiently handles large projects, multiple teams and complex dependencies due to optimized algorithm design and data structures.



Accuracy & Reliability

Accurate priority calculation, dependency resolution and task assignment lead to reliable sprint outcomes and improved team productivity.



Insights & Decision Making

Analytics and visualizations generated using sorting and graph algorithms help stakeholders make data-driven decisions effectively.



Real-world Impact

NexusFlow improves planning accuracy, reduces manual effort, optimizes resource utilization and enhances overall project delivery success.



This project demonstrates the practical application of Design and Analysis of Algorithms in solving real-world Agile project management challenges effectively.

Conclusions

The proposed system demonstrates an intelligent and AI-powered solution for Agile project management by integrating Design and Analysis of Algorithms (DAA) with modern software development practices. By combining Artificial Intelligence with algorithms such as Greedy, Topological Sort, 0/1 Knapsack, Branch and Bound, Merge Sort, Boyer-Moore-Horspool, and Breadth-First Search (BFS), the system automates task prioritization, dependency management, sprint optimization, and resource allocation. It provides optimized project planning, improves team collaboration, enhances decision-making, and offers an efficient, scalable, and reliable solution for modern Agile software development.

References

- [1] Introduction to Algorithms, Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein, *Introduction to Algorithms*, 4th Edition, MIT Press, 2022.
- [2] Martin Fowler, *Patterns of Enterprise Application Architecture*, Addison-Wesley, 2002.
- [3] Ken Schwaber and Jeff Sutherland, *The Scrum Guide: The Definitive Guide to Scrum*, 2020.
- [4] Artificial Intelligence: A Modern Approach, Stuart Russell and Peter Norvig, 4th Edition, Pearson, 2021.
- [5] MongoDB Documentation, MongoDB Inc., and Node.js Documentation, OpenJS Foundation.

QR Codes

